

why plastic heat-sinking materials are highly sought after and studied.

Sofull is said to adopt bile capacitance to replace electrolytic capacitor. According to the spokesperson, graphite enjoys better heat dissipation than aluminium. Smart LED lighting can work at 100-240V AC, and is thus globally suitable. It consumes only around 0.09W, saving energy up to more than 95 per cent for up to 50,000 hours.

Given the developments, how much are LED lighting prices expected to fall each year? For consumer applications, about 20 per cent price decline per year is needed to achieve mass volume by 2015. The LED price will go down 10-20 per cent each year. The focus for the industry is to increase lumens/watt and lumens/\$ for LED lighting. The estimated percentage is around 20 per cent, but it may vary depending on the applications.

Luminous efficacy of 200 lm/W?

The possibility of the luminous efficacy to approach 200 lm/W in the future is not ruled out. According to Sofull, the luminous efficacy is expected to be up to 200 lm/W by 2016. In fact, Cree claims to have already achieved 200 lm/W in its lab in 2011.

The OSRAM spokesperson points out that it really depends on the chip size and driving current applied to the chips. For example, 200 lm/W is a physical limit for a 1mm² chip. However, it may vary for different chip sizes.

The Everlight spokesperson adds that 200 lm/W has already been made possible in some R&D trials. Efficacy may not even be a key topic in the next few years. It could well be colour quality at the forefront of the development focus.

With regard to improving the quality of products, the Sofull spokesperson informs that their company has advanced production equipment. They are providing good workmanship, and exhibiting a smart production process to improve the quality of LEDs.

OSRAM has a very robust new



COB downlight 5W-MS-H01 from Shenzhen Sofull

product qualification process in place in the R&D and quality department, and is targeting to control the DPPM under 0.2 per cent per year.

As for the products having the highest luminous efficacy today, Everlight currently has products in mass production that are 130 lm/W. These are low- to mid-power LEDs for consumer applications and tube lighting. Sofull adds that LEDs from Cree and Nichia have the highest luminous efficacy.

OSRAM claims that most of its SSL products—OSLON SSL, Square and DURIS E—can achieve 130-140 lm/W, depending on the current density per chip area.

To make LEDs that can be more easily used with mainstream electric grids, Everlight is coming up with high-voltage COB LEDs. Sofull is engaged in developing the driver suitable for mainstream power grids. OSRAM is said to be standardising the forward current and voltage to meet the mainstream driver design.

Most LEDs work on voltages that are radically different from that of the mains electricity, resulting in a

significant loss in efficiency. How are makers facing this challenge?

Everlight is using not only high-voltage chips in packages but also multiple DC chips in series in COB packages. On the other hand, Sofull is developing high-conversion-efficiency drivers. At the moment, the engineers have to improve on the features for high-voltage LEDs, which are not yet successful enough in terms of safety and stability.

According to the OSRAM spokesperson, high-voltage

LED is one of the solutions to increase the efficiency of the driver. However, there are certain issues like the safety issue if the LED is driven in high-voltage status.

Package manufacturing process and other development efforts

It would be pertinent to have a look at the LED package manufacturing processes. Everlight is said to be using COB. Now COB is a key focus in the company's product strategy. It requires more labour within a single package, but this saves the labour of mounting multiple discrete components at the fixture assembly.

Sofull uses COB as well. The package manufacturing process includes die bonding, wiring soldering, packaging, baking, etc. OSRAM uses different types of package technologies, e.g., ceramic, plastic lead frame, and silicone, for products. COB packages are now available, e.g., SOLERIQ E.

Also, there are efforts to increase the luminous flux performance and decrease the volume of the device.